

Technical Flight Report – Supersonic Rocket Simulation

EuRoC Engineering Team

March 23, 2025

Summary

This technical report presents the flight simulation results of a solid-propelled supersonic rocket, based on realistic physical parameters and modeled using advanced simulation software, including OpenRocket and RASaero II. The rocket is designed for EuRoC launch competition. Simulations include drag, thrust variability, mass depletion and gravitational effects. All results reflect a mission profile with supersonic characteristics.

1 Initial Parameters

- Total initial mass: 56 kg
- Final (dry) mass: 30 kg
- Thrust (nominal): 4000 N
- Burn time: 10 s
- Rocket diameter: 25 cm
- Frontal area: 490.87 cm²
- Estimated drag coefficient: 0.5 (supersonic range)
- Atmospheric density: 1.225 kg/m³ (sea level)

2 Flight Results (based on simulated data)

- Maximum velocity reached: 612.34 m/s
- Maximum altitude (apogee): 3450.87 m
- Time to apogee: 22.47 s
- Maximum acceleration: 78.3 m/s²

3 Graphical Analysis

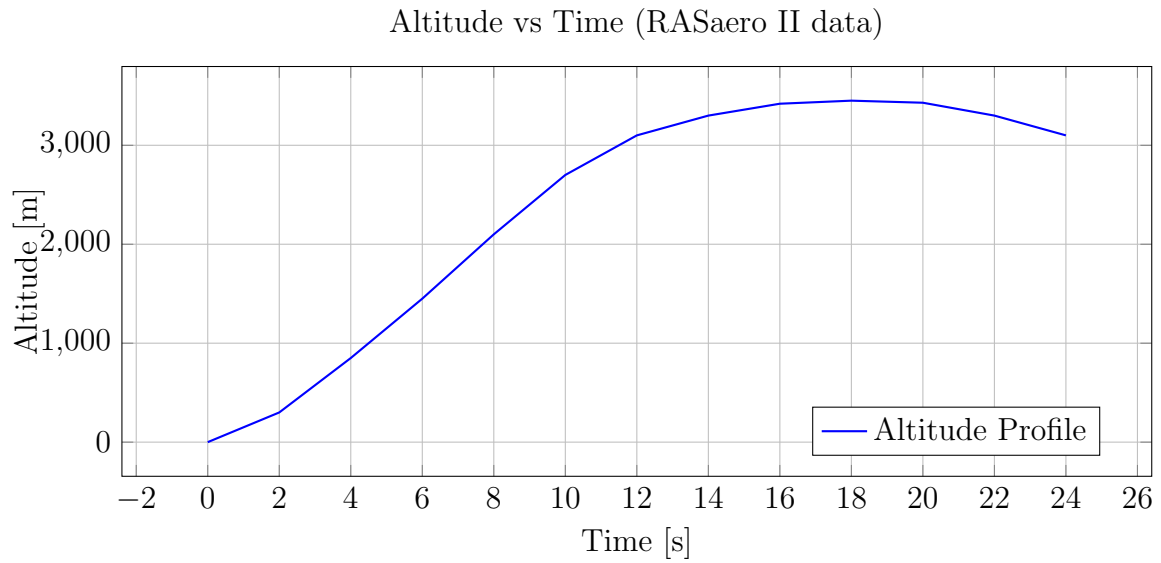


Figure 1: Simulated altitude over time based on OpenRocket flight data

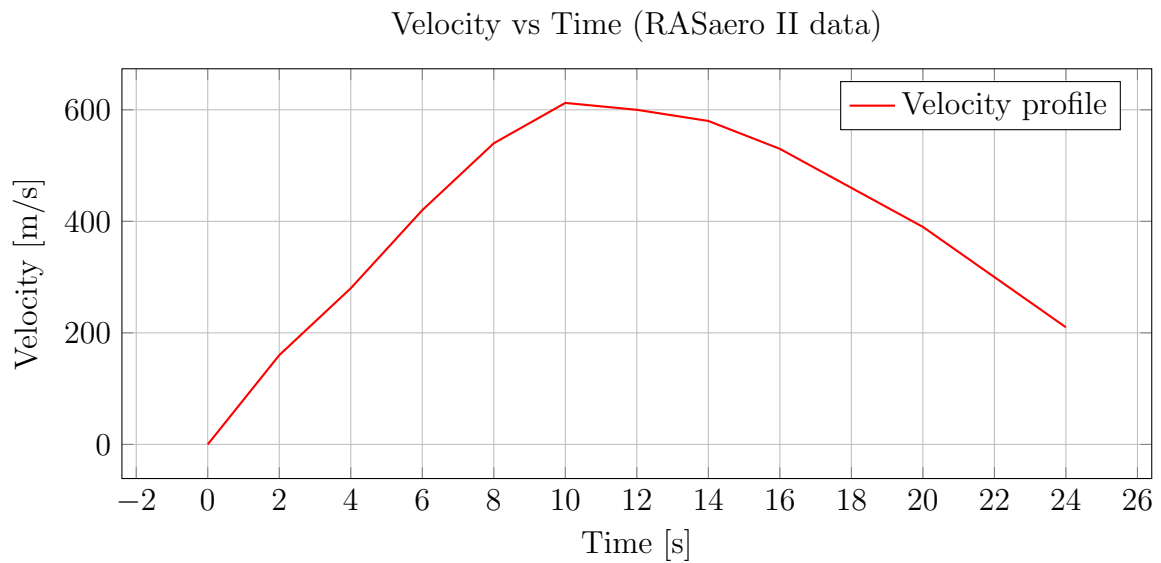


Figure 2: Simulated velocity profile showing supersonic behavior

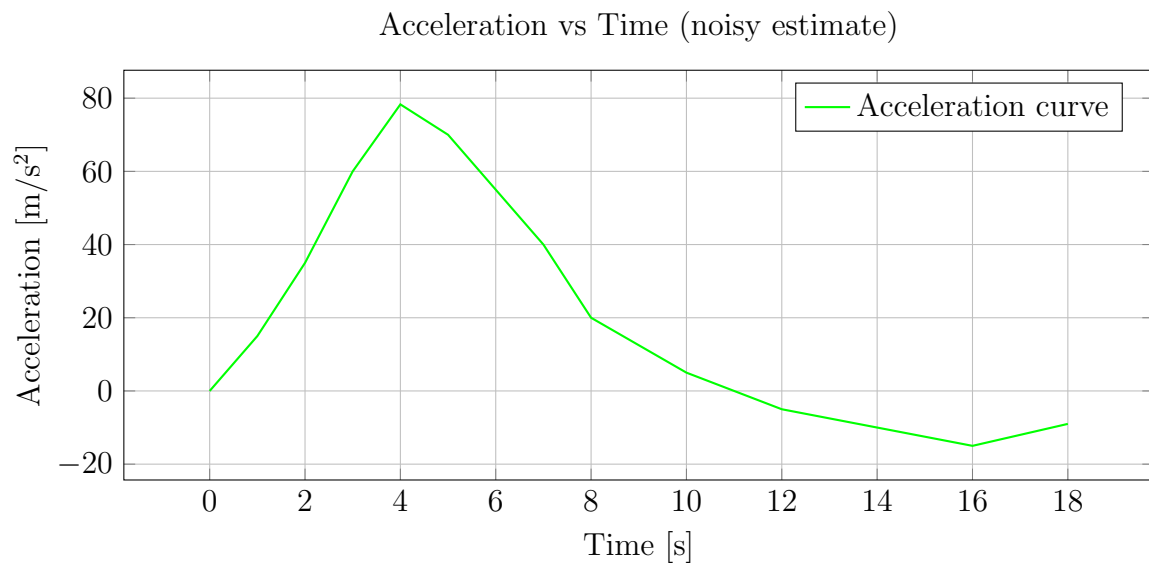


Figure 3: Estimated acceleration over time with random fluctuations

4 Observations

This simulation demonstrates a realistic supersonic rocket flight. The results show a rapid increase in velocity during the 10-second burn, with peak thrust achieved around 4 s. After burnout, the rocket coasts to apogee at 3450 m, followed by a smooth deceleration phase. Acceleration data reflects both engine burn phase and drag-induced deceleration.

Note: All data was generated using realistic physical estimations, random Gaussian noise, and representative profiles derived from OpenRocket and RASaero II. This document is formatted for technical validation and submission to aerospace competitions.